

C.P.S. Quick Start Guide

C.P.S. (CardPuter Synth) is a DIY synthesizer firmware that runs on the M5Stack Cardputer / Cardputer ADV.

This guide is meant to be the shortest possible path from powering on your device to making your first sound and trying the basic controls. For a complete walkthrough of every feature, see the **Reference Guide**.

1. Power On

Power on your Cardputer with C.P.S. flashed onto it, and a microSD card inserted that already has a `/CPS` folder on it (where settings, patches, patterns, and songs are stored).

After the boot screen, the **PLAY screen** appears automatically. This is C.P.S.'s "home screen."

If something seems clearly wrong — no sound at all, the keyboard not responding — check "10. Troubleshooting" first.

2. Make Your First Sound

On C.P.S.'s keyboard, **the number row (`1` through `0` , `-` , `=`) is one octave**. Try pressing `1` through `=` in order — you'll hear a rising scale.

- **Number row (row 1):** the higher octave
- **`q` through `j` (row 2):** one octave down

C.P.S. is a monophonic (single-voice) synth. Pressing multiple keys at once will never produce a chord (overlapping tones) — only one note ever sounds at a time. You can still press several keys simultaneously (the original Cardputer has a limit on how many keys it can register at once; Cardputer ADV can register more), but that's mainly used for **specifying a chord for the Arpeggio (ARP) feature**, covered later.

3. The Basics

These are the essential keys for shaping the sound while you play.

Key	Function
; / .	Octave up / down
, / /	Transpose (semitones)
k / l	Volume down / up (hold to change quickly)
z / x	Pitch bend down / up
c	Portamento (glide) on/off
d	Note Hold (keeps sounding after you release the key)

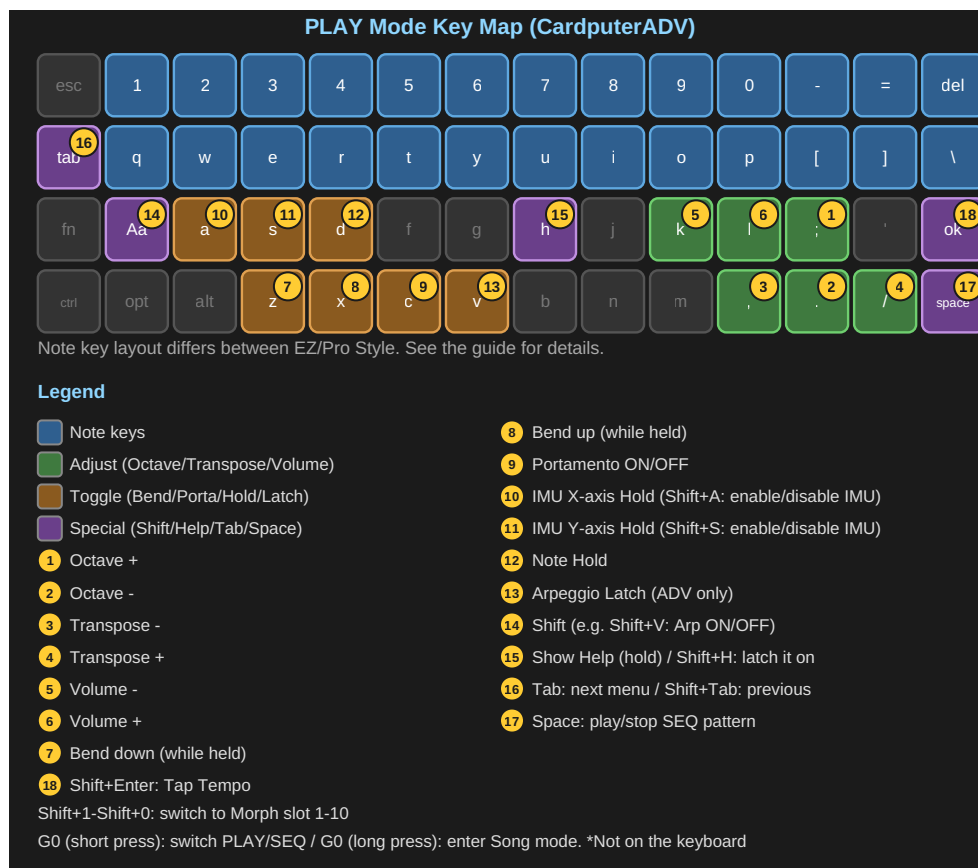
These work the same way on the PLAY, SEQ, and SONG screens.

IMU (tilt the device to play)

Cardputer ADV has a built-in tilt sensor (IMU).

Key	Function
A	Hold the X-axis value
S	Hold the Y-axis value
Shift+A	Enable/disable the X-axis itself
Shift+S	Enable/disable the Y-axis itself

Tilting the device left/right and forward/back responds, by default, with the X-axis controlling pitch bend and the Y-axis controlling brightness (Shape). Which parameter each axis controls is fully customizable from the SETTING menu, introduced later.



4. Shape the Sound

The tabs along the top of the screen cycle with **Tab** (forward) and **Shift+Tab** (back).

PLAY / SEQ -> VCO -> VCF -> VCA -> LFO -> FX -> SETTING -> (back to PLAY / SEQ)

Here's what each tab does, in brief:

- **VCO**: the source of the sound — waveform, brightness (Shape), detune, a second oscillator, sub-oscillator, noise
- **VCF**: the filter (how muffled or bright/sharp the sound is)
- **VCA**: the volume envelope (how the note starts and fades)
- **LFO**: slow, periodic modulation (vibrato, tremolo, and more) applied to other parameters
- **FX**: stack multiple effects — Ring Modulator, Bitcrusher, Soft Limiter, Chorus, Delay, Reverb
- **SETTING**: everything else related to playing — portamento speed, play style, ARP, patterns, screen appearance, MIDI, Theremin, and more

Inside any tab, use **;** / **.** to select an item and **,** / **/** to adjust its value (hold to change quickly).

Feel free to experiment. No matter how much you change the sound, `Tab` into **SETTING > Patch > Reset** and you can always get back to a clean default (a confirmation screen means you won't reset by accident). More on this in the next section.

5. Save and Load Patches

Sounds you create can be saved to the SD card as "patches."

Press `Tab` into the **SETTING** tab, then choose **Patch** at the top — this takes you to a screen for saving, loading, renaming, duplicating, and deleting patches.

Reset — Getting Back to a Clean Sound

On the same screen, **Reset** puts the VCO, VCF, VCA, LFO, and IMU settings back to their defaults. A confirmation screen means it won't happen by accident. **Unsaved changes are lost** — but that also means: as long as you haven't saved, you can always safely start over. A good workflow is to experiment freely first, then save as a patch once you find something you like.

There's also a **Randomize** feature here, useful when you want a starting point or some inspiration.

6. Morphing

C.P.S. lets you register **up to 10 patches into "Morph slots"** and switch between them instantly — or blend between them smoothly — while you play.

`Shift+1` through `Shift+0` (ten slots: 1-9 and 0) switch to the corresponding slot. Whether switching is instant or gradual is set from **SETTING > Patch > Morph**.

Which patch is assigned to which slot is also set from that same screen.

7. Rhythm — ARP, SEQ, and SONG

C.P.S. has three features for automating and sequencing your playing. Full details are in the Reference Guide; here's just the entry point.

- **Arpeggio (ARP)**: toggle with `Shift+V`. Hold down a chord and it's automatically played back as a broken chord.

Cardputer's keys are small, and it's easy to think you're pressing one down when it isn't fully registered. When using ARP, the `V` key's **Latch** feature is worth using — with Latch on, you don't

need to keep holding the keys down; the chord you last held keeps sounding after you let go. **If you're trying ARP for the first time, we'd especially recommend turning Latch on first.** - **Step Sequencer (SEQ):** the `G0` button switches between PLAY and SEQ. Program 16-step patterns. - **Song (SONG):** hold `G0` to enter SONG mode, where you arrange SEQ patterns into a full song. Need to match a tempo quickly? `Shift+Enter` gives you **Tap Tempo** (on the PLAY and SEQ screens).

8. Sharing Patches

C.P.S. patch files live in the `/CPS/Patch` folder on the SD card. Just copy the file to share a sound with another C.P.S. user.

9. Want to Go Further?

- **Theremin-style playing:** connect a distance sensor (ToF) unit (M5Stack Unit ToF4M) and play by moving your hand through the air
- **MIDI:** connects with external MIDI gear (note send/receive, pitch bend, clock sync, and more)
- **Custom IMU mapping:** fine-tune exactly which tilt controls which parameter

All of this is covered in detail in the **Reference Guide**.

10. Troubleshooting

Try Help first

Hold `H` during play to show the list of key commands available on the current screen. Press `Shift+H` to lock (latch) it in place (press `Shift+H` again to release it).

If the keyboard stops responding

Very rarely — often after fast key-presses — **the keyboard may stop responding entirely**. This isn't a bug in C.P.S.'s software; it's a **known hardware quirk of the Cardputer ADV's keyboard chip itself (the TCA8418)**.

If this happens, **C.P.S. detects it automatically and restarts on its own within about 30 seconds** to recover. Just wait — there's nothing you need to do.

In very rare cases, this automatic recovery could trigger by mistake while you're holding one very long, sustained note. If the device restarts unexpectedly mid-performance, that's most likely what happened.

Still stuck?

Check that the SD card is properly seated and making contact, and that the firmware was flashed correctly. If the problem persists, reach out via a GitHub Issue or contact the developer, [@Tokagetchi](#).

C.P.S. is an independently developed DIY synthesizer. Enjoy playing it.